

Monotone upper bounds on non-monotone instances

Violation rates of published monotone upper bounds (M1–M3) and of two ablations of the dual bound (A1, A2) on the experiment instances; the controls never violate.

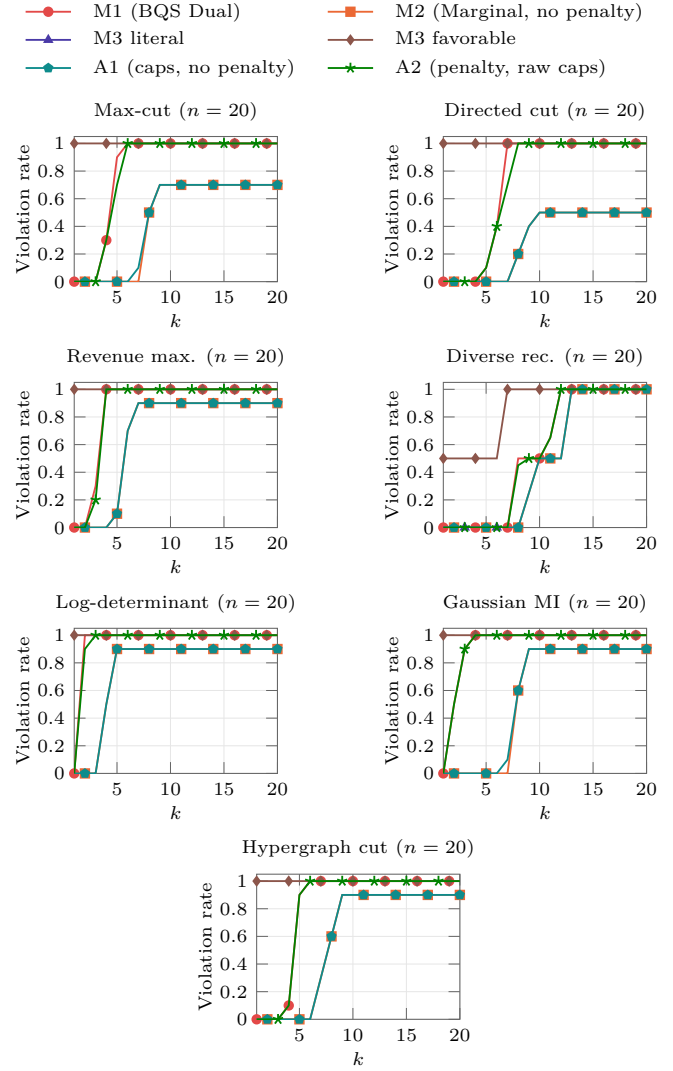


Figure 1: Fraction of rows ($n = 20$ instances, all seeds, OPT by brute force) on which a published monotone upper bound, or an ablation of the dual bound, falls below OPT_k , as a function of k . M1: BQS Dual; M2: Marginal without penalty; M3: Zhang–Tang–Tang Λ^{3*} with non-negative (literal) or free (favorable) increments, rates over the rows where its LP is feasible; A1: the dual bound without its penalty; A2: the dual bound with the raw prefix caps $f_S(A_i)$. The controls (top- k , Marginal, dual bound, B2, B5) never violate.

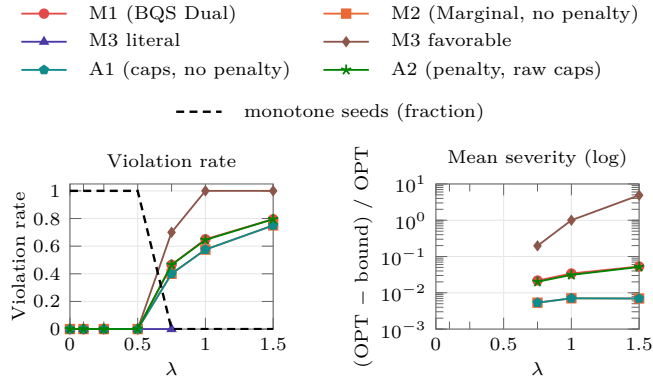


Figure 2: Diverse recommendation with $n = 20$ (10 seeds, $k = 1, \dots, 20$, OPT by brute force) as λ varies: f is monotone for $\lambda \leq 1/2$. Left: fraction of rows on which each method falls below OPT_k (dashed: fraction of monotone seeds). Right: mean relative shortfall over the violated rows (log scale; not shown where a method has no violation). M3-literal is over the rows where its LP is feasible.

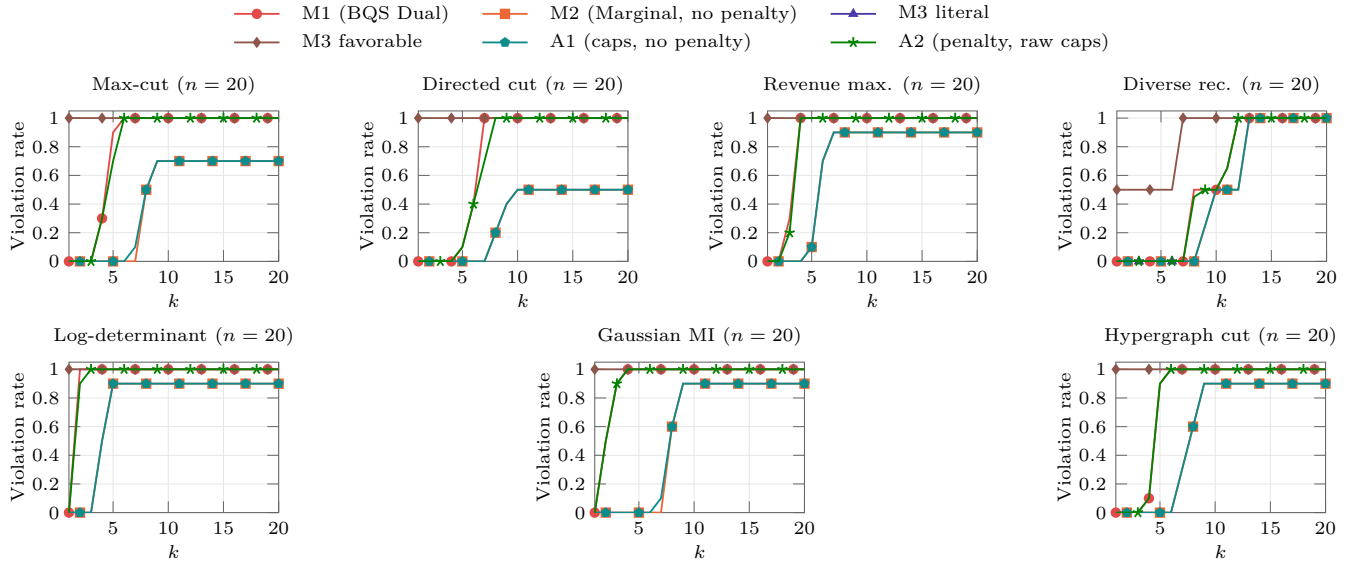


Figure 3: Fraction of rows ($n = 20$ instances, all seeds, OPT by brute force) on which a published monotone upper bound, or an ablation of the dual bound, falls below OPT_k , as a function of k . M1: BQS Dual; M2: Marginal without penalty; M3: Zhang–Tang–Tang Λ^{3*} with nonnegative (literal) or free (favorable) increments, rates over the rows where its LP is feasible; A1: the dual bound without its penalty; A2: the dual bound with the raw prefix caps $f_S(A_i)$. The controls (top- k , Marginal, dual bound, B2, B5) never violate.